RepIt – Feature List (v1)

Hardware Features (Per Device)

Core Components

- Low-power BLE SoC (final selection pending recommendation)
- 6-axis IMU (accelerometer + gyroscope)
- Rechargeable LiPo battery
- Battery charging circuit (USB-C preferred)
- Battery protection + fuel monitoring (if feasible in v1)
- Status LED indicator
- Single user button (power / pairing / reset)
- On-board voltage regulation
- Attachment area for strap

Dual-Device System Features

- Two identical hardware units
- Event-based synchronization between devices
- Timestamp alignment for motion data
- Role-based architecture (temporary master + peer)
- Automatic reconnection logic
- Independent BLE connection from each device to mobile app

Communication Features

- BLE 5.x support
- Stable data streaming under moderate activity
- Configurable connection intervals
- Low-power sleep modes
- Structured packet design for motion frames

Power Requirements

- Minimum full-day active runtime
- IMU interrupt-based wake strategy
- Deep sleep when inactive
- Optimized duty cycling

Firmware Architecture (Target Structure)

- Hardware abstraction layer
- Power management layer
- Communication layer (BLE + sync logic)
- Application layer (data acquisition + formatting)

OTA support desirable (if timeline permits).

Mechanical / Enclosure (Prototype Phase)

- Compact wearable form factor
- 3D printable enclosure (first iteration)
- LED visibility window
- Button accessibility
- Internal layout considering antenna clearance and battery cavity

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Data & ML Integration

- Continuous motion capture during active sessions
- Windowed data transmission to mobile app
- Initial ML inference will run on mobile
- Future potential for TinyML (not required in first build)

Timeline Targets

- Week 3: Functional synchronized prototype ready for 3D print + ML testing
- Week 6: Refined PCB + production-ready foundation